

Two TEEP Student Research Projects: AI for Physiological Signals & Aging-in-Place Monitoring

Prepared for: Faculty supervisor (AI/ML, Biomedical Signal Processing, Gerontechnology) **Students:** 2 TEEP students, Computer Science background, no prior biomedical research experience **Duration:** 6–12 months, no IRB, no patient data collection, public datasets only

How These Two Projects Were Selected

I searched current literature (2022–2026) and dataset repositories (PhysioNet, UCI ML Repository, PubMed, arXiv, ScienceDirect, Frontiers, Nature Scientific Data) against your five priorities: **dataset accessibility, label availability, feasibility, research novelty, and publication potential**, weighted toward alignment with your program (physiological AI, explainable/trustworthy health AI, aging-in-place monitoring). Several attractive-sounding topics were **rejected** before finalizing these two — the reasons are documented in the "Topics Considered and Rejected" section near the end, since you asked me to explain rejections explicitly rather than just omit them.

The two finalized projects are:

Project A — Age-Stratified Explainable & Uncertainty-Calibrated ECG Diagnosis (classification, PTB-XL)

Project B — Subject-Independent, Personalizable Cuffless Blood Pressure Estimation Across Clinical and Free-Living Settings (regression, PulseDB + PPG-DaLiA)

They are deliberately different in **signal type (ECG vs. PPG)**, **task type (multi-label classification vs. regression)**, **core ML technique (uncertainty quantification/explainability vs. domain adaptation/personalization)**, and **dataset ecosystem** — so the two students will not overlap or compete for the same data/compute pipeline, while both plug directly into your aging/physiological-AI research line.

PROJECT A — Age-Stratified Explainable & Uncertainty-Calibrated ECG Diagnosis

A. Project Title

"Reliable and Interpretable Multi-Label ECG Diagnosis Across the Age Spectrum: A Conformal-Prediction Approach to Trustworthy Cardiac Screening in Older Adults"

B. Research Question

Does a lightweight ECG classifier's diagnostic reliability (calibration, uncertainty, and explanation fidelity) systematically degrade for older patients, and can distribution-free uncertainty quantification (conformal prediction) combined with saliency-based explanation be used to detect and flag those unreliable predictions specifically in the elderly subgroup, without sacrificing overall accuracy?

C. Hypothesis

Standard subject-independent ECG classifiers (trained without age-aware calibration) will show measurably wider performance gaps and lower calibration quality for patients aged 65+ compared to younger patients — consistent with recent findings that age is one of the strongest sources of demographic performance disparity in ECG deep learning<cite index="48-1,49-1">bias was defined as significant differences across demographic subgroups, and a large retrospective study found the performance of an ECG deep learning model predicting heart failure declined with age</cite>. We hypothesize that adding a **conformal-prediction layer** on top of a standard classifier will produce prediction sets whose *size* (not just point accuracy) reliably signals this age-related unreliability — giving clinicians an actionable "this prediction needs a human look" flag specifically where it matters most for elderly screening — while a lightweight explainability layer (Grad-CAM / SHAP on lead-wise attributions) will show *where* the model's attention shifts for less-confident elderly cases.

D. Research Gap

Explainability and uncertainty quantification for ECG deep learning are each individually well studied (SHAP/Grad-CAM interpretability papers on PTB-XL are common, and conformal prediction is an active general ML topic), but:

Most PTB-XL explainability papers report a single aggregate accuracy/F1 number and do not stratify performance, calibration, or explanation quality **by age subgroup**, even though PTB-XL's own age range (0–95, median 62) makes this analysis directly possible without any new data collection.

Demographic-disparity studies on ECG deep learning exist (age/race/sex bias in heart-failure prediction) but they use **large private hospital datasets** and do not combine disparity analysis with **distribution-free uncertainty quantification** or **explanation-level auditing**.

Conformal prediction for medical time series is an emerging but still small literature; combining it specifically with **age-subgroup auditing** for ECG has not been done on an open, reproducible benchmark like PTB-XL. This is exactly the kind of "small but meaningful contribution" you asked for: it does not propose a new architecture, but combines two established but rarely-combined trustworthy-AI tools (conformal prediction + explainability) around a genuinely open and clinically important question (does ECG AI quietly fail older patients?) — which is a natural fit for your gerontechnology/aging research line.

E. Proposed Contribution

The student will:

Train and subject-stratified-validate a multi-label ECG classifier (baseline CNN/ResNet1D) on PTB-XL.

Stratify all evaluation metrics by age decade/group (e.g., <40, 40–65, 65–80, 80+), quantifying the performance and calibration gap.

Implement class-conditional conformal prediction on top of the trained classifier to produce statistically guaranteed prediction sets, and test whether **prediction-set size** is a useful, learned-for-free early warning signal of age-related unreliability.

Apply Grad-CAM/SHAP explanation and quantify whether explanation stability/consistency also degrades for the same low-confidence elderly cases (a novel triangulation of three trustworthiness signals — accuracy, uncertainty, explanation — around one demographic axis).

Package it into a small, reusable “age-fairness audit” evaluation pipeline that could plausibly generalize to your future elderly cardiac monitoring work.

F. Dataset

Field	Detail
Dataset name	PTB-XL
Official source	PhysioNet
URL	https://physionet.org/content/ptb-xl/1.0.3/
Registration required	No credentialing/CITI training required — open-access dataset released under CC-BY 4.0 , directly downloadable
Data-use agreement	No formal DUA; standard open-data attribution license only
Immediate download	Yes — <code>wget -r -N -c -np https://physionet.org/files/ptb-xl/1.0.3/</code> works immediately, no approval wait
Signal types	12-lead ECG (I, II, III, aVR, aVL, aVF, V1-V6)
Sampling frequency	Provided at both 100 Hz and 500 Hz versions
Subjects / records	21,837 records from 18,885 patients, age range 0–95 (median 62)
Data size	~two vendor formats (WFDB), roughly 2–3 GB (100 Hz version) to ~20 GB (500 Hz), easily Colab/workstation-feasible at 100 Hz
Labels	Cardiologist-annotated SCP-ECG statements: 71 individual statements → 5 diagnostic superclasses (NORM, MI, STTC, CD, HYP) and rhythm/form sub-statements; multi-label per record
Label granularity	Record-level (per 10-second recording)
Demographics available	Age, sex, height, weight — included directly in metadata, exactly what’s needed for age-stratified analysis

G. Label Verification (Mandatory)

The ground-truth diagnostic labels come directly from the `scp_codes` field of PTB-XL’s `ptbx1_database.csv` metadata file, which encodes each recording’s cardiologist-assigned SCP-ECG statement(s) with an associated confidence percentage (0–100%) reflecting inter-annotator likelihood. These are mapped via the

officially provided `scp_statements.csv` lookup table to the standard **5-superclass hierarchy** (NORM/MI/STTC/CD/HYP), which is the community-standard target used in essentially all PTB-XL benchmark papers, enabling direct baseline comparison. Age and sex fields sit in the same metadata row per `ecg_id`, so **no additional linking, inference, or manual annotation is required** — age-stratified evaluation is a pure re-slicing of existing, already-verified ground truth. The dataset also ships an official recommended stratified 10-fold cross-validation split (`strat_fold` column) built to preserve class balance across folds, which the student will use as the basis for a subject-independent split (see Section L).

Limitation to disclose: SCP-ECG labels are diagnostic categories, not explicit "ground truth risk" — they reflect what cardiologists documented on that ECG, and confidence values below 100% indicate genuine clinical ambiguity for some labels. This is a normal and accepted limitation in all PTB-XL literature, not specific to this project.

H. Proposed Method — Pipeline



Raw 12-lead WFDB signals (100 Hz)

- Preprocessing: bandpass filter (0.5–40 Hz), baseline wander removal, per-lead z-normalization
- Segmentation: fixed 10-second windows (already record-aligned; no further segmentation needed)
- Feature/representation: raw signal fed to 1D-CNN backbone (no hand-crafted features needed)
- Baseline models: trained and subject-independently validated
- Proposed model: baseline + conformal prediction wrapper + Grad-CAM/SHAP explanation module
- Evaluation: stratify all metrics (below) by age subgroup; compare baseline vs. proposed

I. Baseline Models

Random Forest on hand-crafted features (HRV, QRS morphology, wavelet features) — classical, fast, sanity-check baseline.

1D-CNN (e.g., simple 4–6 layer convolutional network) — standard PTB-XL baseline in the literature.

ResNet1D — the most common strong deep baseline reported for PTB-XL superclass classification.

(*Optional stretch*) a small Transformer/attention-based 1D model, only if time allows after core deliverables are done.

J. Proposed AI Method

One main improvement, as requested: add a **class-conditional (Mondrian) conformal prediction layer** on top of the best baseline, calibrated per age subgroup, to produce statistically valid prediction sets with a user-specified coverage guarantee (e.g., 90%). This is the single novel technical addition — explainability (Grad-CAM/SHAP) is used as an *analysis tool* around it, not as a second competing "method," keeping the project appropriately scoped for one student.

K. Evaluation

Metric	Role
AUROC / AUPRC (per class, per age subgroup)	Primary — standard for imbalanced multi-label ECG tasks (PTB-XL classes are imbalanced, e.g., HYP ~12%), AUPRC especially important for rare/minority classes
F1-score (macro & per-class)	Primary — standard PTB-XL comparison metric
Sensitivity / Specificity	Secondary — clinically interpretable, especially per age group
Cohen's kappa	Secondary — agreement/reliability check
Conformal coverage & average prediction-set size (per age subgroup)	Primary novel metric — this is the core deliverable: does set size reliably grow for elderly patients where the model is less reliable?

AUROC/AUPRC and F1 are primary because they are the field-standard PTB-XL comparison numbers (needed to show the method doesn't hurt baseline accuracy); conformal set size stratified by age is the *primary novelty metric* because it is the actual research question.

L. Experimental Design

Subject-level split: Use PTB-XL's official `strat_fold` (1–10), reserving folds 9–10 as held-out test (as recommended in the original paper) — this is already subject-disjoint, preventing leakage of the same patient across train/test.

Cross-validation: 8-fold training/validation within folds 1–8 for hyperparameter tuning.

Class imbalance handling: class-weighted loss / focal loss for minority classes (HYP is the smallest class); report per-class AUPRC to avoid imbalance masking.

Conformal calibration set: a held-out calibration split (distinct from test) used purely to compute conformal thresholds, standard conformal-prediction protocol — must not overlap with test subjects.

Ablation experiments: (1) baseline vs. + conformal only, (2) global conformal calibration vs. age-conditional (Mondrian) calibration, (3) with/without explanation-consistency filtering.

External validation (optional stretch): MIMIC-IV-ECG (also PhysioNet, though larger and requires simple credentialed access) or the Chapman-Shaoxing dataset (also open) could serve as an external cross-dataset check if time allows — not required for the core deliverable.

M. Expected Difficulty

Dimension	Rating
Dataset acquisition	Easy (direct download, no credentialing)
Data preprocessing	Easy-Moderate (WFDB format is well documented, official Python loading code exists)
AI implementation	Moderate (baselines are standard; conformal prediction adds a learnable but well-documented statistical layer)
Biomedical knowledge required	Low-Medium (label taxonomy needs to be learned but is documented; no deep cardiology expertise needed)
Computing requirement	Low-Medium (Colab T4/A100 or a normal GPU workstation is sufficient at 100 Hz)
Research novelty	Medium-High
Publication potential	Medium-High

N. 6-12 Month Timeline

Month	Milestone
1	Literature review (PTB-XL benchmarks, conformal prediction basics, ECG demographic bias papers); set up Colab/workstation environment
2	Download PTB-XL, build preprocessing + data-loading pipeline, reproduce official train/val/test split
3	Implement and train baseline 1D-CNN and ResNet1D; reproduce literature-comparable F1/AUROC
4	Age-stratify evaluation of baselines; quantify the raw performance/calibration gap (this alone is a publishable finding even in isolation)
5	Implement conformal prediction wrapper (global calibration first, then age-conditional/Mondrian calibration)
6	Evaluate conformal set sizes and coverage per age subgroup; tune calibration
7	Add Grad-CAM/SHAP explanation module; qualitative + quantitative explanation-consistency analysis for low-confidence elderly cases
8	Ablation studies (global vs. age-conditional calibration; class-imbalance handling variants)
9	(Optional) external/cross-dataset validation on MIMIC-IV-ECG or Chapman-Shaoxing
10	Consolidate results, generate all figures/tables, sensitivity analysis
11	Draft manuscript / conference paper
12	Revise, internal review, submit

O. Risks and Backup Plan

Risk	Mitigation / Backup
Baseline underperforms published PTB-XL numbers	Use the smaller, well-established 5-superclass task (not all 71 labels) as the core deliverable — this is the most reproducible target in the literature
Age-related performance gap turns out to be small/insignificant	Still publishable as a "reassuring" negative/neutral finding; pivot emphasis to whether conformal set size is <i>still</i> a useful general reliability signal (not necessarily age-specific)
Conformal prediction implementation issues	Well-supported open-source libraries exist (e.g., MAPIE, <code>crepes</code> , <code>puncc</code>) — student does not need to implement conformal theory from scratch
Insufficient elderly (80+) samples for robust subgroup statistics	PTB-XL age range goes to 95 with substantial 65+ representation; if 80+ bin is too sparse, use 65+ as primary elderly cutoff and treat 80+ as a smaller exploratory analysis
Computing limitations	100 Hz version of PTB-XL is small enough for Colab free tier; no need for the 500 Hz version

P. Publication Potential

Suitable venues: IEEE EMBC, IEEE-EMBS BHI, Computing in Cardiology, ML4H (Machine Learning for Health) workshop, or a biomedical informatics journal (e.g., *JBHI*, *Physiological Measurement*).

Minimum publishable result: a rigorous, reproducible demonstration that (a) an age-related reliability gap exists (or rigorously does not) in a standard open ECG benchmark, and (b) a distribution-free, statistically grounded reliability signal (conformal set size) tracks it — even without inventing a new mitigation method, this kind of careful, well-controlled auditing study is publishable at a workshop or applied-ML health venue.

PROJECT B — Subject-Independent, Personalizable Cuffless Blood Pressure Estimation

A. Project Title

"From ICU to Everyday Life: Subject-Independent Evaluation and Few-Shot Personalization for Cuffless PPG-Based Blood Pressure Estimation Under Domain Shift"

B. Research Question

How much does cuffless PPG-based blood pressure estimation performance degrade when evaluated under a **rigorous, subject-independent, cross-setting protocol** (train on ICU data, test on free-living wearable data) compared to the commonly reported but leakage-prone in-distribution evaluation protocol — and how

much of that gap can be recovered using a lightweight, few-shot personalization/calibration step suitable for elderly home-monitoring users?

C. Hypothesis

We hypothesize (a) that reported cuffless BP estimation accuracy in much of the literature is substantially **inflated by record-level (not subject-level) data leakage and by evaluating exclusively on in-distribution ICU data**; and (b) that a brief per-subject calibration step (a small number of labeled calibration segments, simulating what a home user could feasibly provide, e.g., during a single cuff-based validation visit) will recover a clinically meaningful fraction of that accuracy loss when transferring the model to a free-living wearable setting — a scenario directly relevant to non-intrusive, remote elderly BP monitoring.

D. Research Gap

This is a rare case in this literature where the **research gap is a documented methodological flaw**, which makes the project both feasible and clearly novel:

The most commonly used cuffless BP dataset (the UCI "Cuff-Less Blood Pressure Estimation" set, derived from MIMIC-II) does **not include subject IDs**, so [it has the issue of suppressing patient identification information, making it impossible to track whether two records belong to the same subject, which can lead to data leakage in machine learning applications](#). Many published papers unknowingly train and test on data from the same patients.

PulseDB, a newer and more rigorously curated dataset, was built specifically to fix this: [it contains over 5.2 million high-quality 10-second segments of ECG, PPG, and arterial blood pressure waveforms from 5,361 subjects, retrieved from the MIMIC-III matched subset and the VitalDB database, and includes subject identification and demographic information that can be used to evaluate model generalizability to unseen subjects](#). It also ships official, reproducible **subject-balanced training / calibration-based testing / calibration-free testing** splits, which is exactly the rigor this project needs. However, even PulseDB is still ICU/OR-derived (MIMIC-III, VitalDB), and BP estimation models are rarely tested for **cross-setting domain shift** into a genuinely free-living wearable context. PulseDB's own authors demonstrated this gap: after training calibration-free on hemodynamically compromised ICU/OR data and testing on a distinct held-out AAMI-compliant subset, error was clinically unacceptable, but [using an initial calibration segment with transfer-learning fine-tuning brought both mean and standard-deviation errors within acceptable clinical limits, and this personalization method significantly outperformed the calibration-free approach](#) — and the same authors flagged **PPG-DaLiA** (a genuinely free-living, wrist-worn daily-activity dataset) as the natural next external benchmark for this kind of transfer.

What a TEEP student can realistically contribute: a clean, reproducible, subject-honest benchmark comparing (1) naive calibration-free cross-setting transfer, (2) rigorous subject-independent PulseDB-only evaluation, and (3) few-shot personalized calibration — with PPG-DaLiA as genuine external/free-living validation. This directly matters for your remote/aging-in-place monitoring research line, since it answers "how much cuff-based validation would an older adult actually need to provide at home for a wearable BP estimate to be trustworthy?"

E. Proposed Contribution

The student will:

Build a clean training/evaluation pipeline on PulseDB's official subject-balanced splits (fixing the leakage problem present in the older, still widely-used UCI dataset).

Establish an honest subject-independent baseline (no calibration) for SBP/DBP estimation from PPG (+ECG features).

Implement a **few-shot personalization** method (lightweight fine-tuning or feature-space calibration using a small number of labeled segments per new subject) and quantify how much error it recovers.

Evaluate the trained model on **PPG-DaLiA** — a free-living, wrist-worn wearable dataset never seen during training — as a genuine domain-shift test, and quantify how much personalization helps in that harder, more realistic transfer.

Deliver a clear "leakage vs. honest evaluation vs. personalized honest evaluation" comparison table that is directly useful for anyone (including your future students) choosing an evaluation protocol for cuffless BP work.

F. Dataset

Field	Primary training/eval dataset	External domain-shift dataset
Name	PulseDB	PPG-DaLiA
Official source	Figshare (Frontiers) / GitHub	UCI Machine Learning Repository
URL	https://doi.org/10.3389/fdgth.2022.1090854 (paper); data + code: GitHub repo linked from paper; Kaggle mirror also provided by authors	https://archive.ics.uci.edu/dataset/495/ppg+dalia
Registration required	No — open download (Figshare/Kaggle), CC BY-NC-SA 4.0 license	No — open, direct UCI download
Data-use agreement	License terms only (non-commercial share-alike), no formal DUA/credentialing	None
Immediate download	Yes	Yes (also mirrored on Zenodo/Kaggle as pickle files)
Signal types	Synchronized ECG, PPG, arterial BP (ABP) waveforms, plus subject demographics	Wrist PPG (Empatica E4) + 3-axis accelerometer; chest ECG + respiration + accelerometer (RespiBAN) — no direct BP labels
Sampling frequency	Standardized 10-s segments (source: 125 Hz MIMIC-III / 100–500 Hz VitalDB, harmonized in PulseDB)	PPG 64 Hz, wrist accel 32 Hz; chest ECG/respiration 700 Hz
Subjects / records	5,361 subjects, 5,245,454 10-second segments	15 subjects, ~64,700 labeled 8-second windows
Data size	Large (~tens of GB full set; manageable subsets/splits provided)	Small (~a few GB), very Colab-friendly
Labels	Continuous SBP/DBP (and full ABP waveform) per segment, subject ID included	Heart rate (from ECG) — used for a secondary, easier validation task , not as a BP label
Label granularity	Segment-level, subject-linked	Window-level

G. Label Verification (Mandatory)

PulseDB (primary dataset): Ground-truth systolic/diastolic BP values are derived directly from the synchronized invasive arterial blood pressure (ABP) waveform recorded alongside PPG/ECG in the source MIMIC-III and VitalDB databases; PulseDB's authors applied signal-quality filtering and peak-detection (via an established open-source ABP pulse-onset detection algorithm) to extract per-segment SBP/DBP directly from these ABP waveforms — this is a physical, instrument-derived ground truth, not an inferred or manually annotated label. Critically, PulseDB retains **subject identifiers**, so the student can construct a strictly subject-disjoint train/test split (unlike the older UCI dataset), directly using the **official subject-balanced training set and calibration-free/calibration-based testing sets already released and reproducible via the authors' GitHub scripts** — no custom leakage-prevention logic needs to be invented from scratch.

PPG-DaLiA (external validation dataset): This dataset does **not** contain BP labels at all — it is used only as an unlabeled, real-world **input-distribution** stress test (does the PPG waveform "look" different enough from ICU PPG that a PulseDB-trained model's predictions become unreliable/out-of-range?), and, separately, as a **fully labeled secondary sanity-check task** for heart-rate estimation, where the ECG-derived heart rate is [the ground-truth reference for the wrist PPG-based HR estimation task, obtained from a chest-worn ECG](#). This dual role is stated explicitly here so it is unambiguous: PPG-DaLiA supports domain-shift *analysis* (waveform distribution shift, physiologically-plausible-range checks, and — if the student wants a fully labeled cross-task check — heart-rate estimation transfer) but does **not** provide independent BP ground truth for direct SBP/DBP accuracy scoring. This is disclosed as an explicit limitation, not glossed over.

H. Proposed Method — Pipeline



Raw synchronized PPG+ECG+ABP segments (PulseDB)

- Preprocessing: bandpass filtering, PPG/ECG fiducial point detection (onset, peak, dicrotic notch)
- Segmentation: use PulseDB's official 10-s segments (already pre-cleaned)
- Feature/representation: pulse-transit-time & PPG-morphology features + raw-waveform CNN branch
- Baseline models: trained + subject-independently validated on PulseDB splits
- Proposed model: baseline + few-shot personalization (per-subject fine-tuning/calibration)
- Evaluation: (1) PulseDB subject-independent test, (2) domain-shift test on PPG-DaLiA

I. Baseline Models

Random Forest / Gradient Boosting on hand-crafted pulse-transit-time and PPG-morphology features — classical, interpretable, fast baseline widely used in this exact literature.

1D-CNN on raw PPG (+ECG) waveform — standard deep baseline.

ResNet1D / lightweight Transformer — stronger deep baseline reported in recent PulseDB-benchmarking papers.

(*Reference-only, not to be re-implemented*) naive UCI-leakage-style evaluation numbers from the literature, quoted only to illustrate the inflation gap, not benchmarked directly by the student.

J. Proposed AI Method

One main improvement, as requested: a **lightweight few-shot personalization** layer — e.g., a small per-subject fine-tuning of the final regression layers (or a feature-space affine recalibration) using only a handful of labeled calibration segments per new subject, inspired by the transfer-learning calibration approach PulseDB's own authors validated. This keeps the technical scope appropriately bounded to one clear method (not combining domain adaptation + fusion + foundation models all at once).

K. Evaluation

Metric	Role
Mean Absolute Error (MAE) for SBP and DBP	Primary — standard clinical BP-estimation metric, directly comparable to AAMI/BHS standards
Standard deviation of error (SDE)	Primary — AAMI protocol explicitly requires both mean error and SD of error within limits
Compliance with AAMI/BHS grading (mean error ≤ 5 mmHg, SD ≤ 8 mmHg)	Primary — the field-standard pass/fail clinical benchmark
Correlation coefficient (Pearson r) between predicted and true BP	Secondary
Bland-Altman agreement plots	Secondary, standard for BP method comparison

MAE and AAMI-standard compliance are primary because they are the metrics the entire cuffless-BP literature uses for comparability, and because AAMI compliance is the actual clinical bar a home-monitoring method would need to meet.

L. Experimental Design

Subject-level split: use PulseDB's **official subject-balanced training set**, with the standard calibration-free test set held out at the subject level — this is built into the dataset release specifically to prevent leakage, so the student uses it rather than building custom splitting logic.

Cross-validation: k-fold (subject-level) within the training subjects for hyperparameter tuning.

Prevention of leakage: explicit assertion checks in code (no shared subject IDs between train/calibration/test splits) — this is itself worth reporting as a methodological contribution, since it directly documents what much prior work skipped.

Class/target imbalance handling: BP value distribution is typically right-skewed in ICU populations (more hypertensive/hypotensive extremes); apply stratified sampling or a weighted loss over BP value bins.

Personalization protocol: for a held-out subject, use only the *first* small % of that subject's segments as "calibration" data (as PulseDB's own calibration-based protocol does) and evaluate on the remainder — never let calibration data leak into the reported test metric.

External/domain-shift validation: PPG-DaLiA, as described above (distributional shift + optional HR-estimation cross-check), which is the most novel design choice differentiating this from typical single-

dataset cuffless-BP papers.

Ablation experiments: (1) calibration-free vs. personalized, (2) amount of calibration data needed (1 vs. 3 vs. 5 calibration segments) — directly answers "how much would an elderly home user need to provide," (3) classical features vs. raw-waveform CNN.

M. Expected Difficulty

Dimension	Rating
Dataset acquisition	Easy-Moderate (PulseDB is larger and requires more careful setup than a single small file, but no credentialing)
Data preprocessing	Moderate (PPG fiducial point detection has some signal-processing learning curve, but well-documented open-source tools like <code>pyPPG</code> / <code>heartpy</code> exist)
AI implementation	Moderate (regression baselines are standard; personalization/fine-tuning loop is a well-documented pattern)
Biomedical knowledge required	Medium (pulse-transit-time physiology and AAMI/BHS BP standards need to be learned, but are well documented)
Computing requirement	Medium (PulseDB is large; recommend using a curated subset — e.g., the VitalDB-only calibration-free/AAMI subsets — rather than the full 5.2M-segment set, to stay Colab/workstation-feasible)
Research novelty	Medium-High
Publication potential	Medium-High

N. 6-12 Month Timeline

Month	Milestone
1	Literature review (cuffless BP estimation, PulseDB paper, AAMI/BHS standards, PTT physiology); environment setup
2	Download PulseDB (recommend starting with the smaller, official VitalDB calibration-free/AAMI subsets rather than the full MIMIC-III+VitalDB set); build preprocessing pipeline
3	Implement feature extraction (PTT, PPG morphology) + baseline Random Forest / Gradient Boosting model
4	Implement 1D-CNN / ResNet1D deep baseline; reproduce subject-independent PulseDB benchmark numbers
5	Quantify and document the "leakage gap": rerun evaluation under a naive (non-subject-disjoint) split purely as a controlled comparison illustrating the inflation problem (small controlled experiment, not the main result)
6	Implement few-shot personalization/calibration method
7	Evaluate personalization on PulseDB held-out subjects (calibration-based protocol); tune calibration-data amount
8	Download and preprocess PPG-DaLiA; run domain-shift evaluation (distribution shift + personalization on this external set)
9	Ablation studies (feature type, amount of calibration data, model architecture)
10	Consolidate results, Bland-Altman plots, AAMI compliance tables
11	Draft manuscript / conference paper
12	Revise, internal review, submit

O. Risks and Backup Plan

Risk	Mitigation / Backup
PulseDB's full size is unwieldy for Colab	Use the authors' pre-built, smaller official subsets (train / calibration-based test / calibration-free test / AAMI test) rather than the full 5.2M-segment release
AAMI-standard error targets are not met even with personalization	Still a valid, publishable negative/quantified finding — the field currently does not reliably meet AAMI targets calibration-free either, so a careful, honest quantification is itself useful
PPG-DaLiA's PPG signal characteristics (wrist, motion-heavy) are too different for any meaningful transfer	Reframe as the core finding itself (quantifying <i>how large</i> the ICU→wearable domain gap is a legitimate and interesting result), and add WESAD (also open, chest+wrist PPG/ECG in a lower-motion protocol) as an intermediate-difficulty domain-shift dataset
Computational cost of large-scale PulseDB training	Subsample to a fixed number of subjects (e.g., 500–1000) while preserving the official subject-disjoint split structure
Motion artifacts dominate PPG-DaLiA signal quality	Use only lower-motion activity segments (e.g., sitting, working) from PPG-DaLiA's activity labels as an easier initial domain-shift test before attempting high-motion segments

P. Publication Potential

Suitable venues: IEEE EMBC, IEEE-EMBS BHI, Computing in Cardiology, *Physiological Measurement*, *Biomedical Signal Processing and Control*, or an ML4H workshop.

Minimum publishable result: a rigorous, subject-honest quantification of (a) the leakage-driven performance inflation gap versus prior UCI-based papers, (b) how much few-shot personalization recovers under honest evaluation, and (c) how far performance degrades under genuine ICU→wearable domain shift. Even without beating any state-of-the-art number, this kind of careful **benchmark correction and calibration study** is directly useful and citable to the field, matching the "expose a flawed evaluation practice + propose a fix" pattern that has been well received in recent methodology-focused papers in this area.

Final Comparison Table

Criteria	Student 1 — Project A	Student 2 — Project B
Research topic	Age-stratified explainable & uncertainty-calibrated ECG diagnosis	Subject-independent, personalizable cuffless BP estimation across domain shift
Dataset	PTB-XL (PhysioNet)	PulseDB (primary) + PPG-DaLiA (external validation)
Dataset accessibility	Excellent — instant open download, no credentialing	Good — open download, larger file size, recommend using official pre-built subsets
Labels available	Yes — cardiologist multi-label diagnostic codes, record-level, directly usable	Yes — instrument-derived continuous SBP/DBP from ABP waveform, segment-level, subject-linked
Main task	Multi-label ECG classification	Continuous BP regression (SBP/DBP)
Main AI method	Class-conditional conformal prediction + explainability audit	Few-shot personalization / calibration under domain shift
Novelty	Medium-High (fairness/reliability auditing angle, underexplored combination)	Medium-High (exposes and corrects a known evaluation flaw in the field)
Difficulty	Moderate	Moderate-Somewhat higher (larger dataset, PPG signal processing curve)
Computing requirement	Low-Medium (Colab-friendly at 100 Hz)	Medium (recommend curated PulseDB subsets)
Expected duration	6-12 months	6-12 months (slightly back-loaded due to domain-shift phase)
Publication potential	Medium-High (workshop/applied ML4H venue realistic)	Medium-High (methodology/benchmark-correction framing is well received)
Relationship to your research	Directly serves explainable AI for healthcare + AI for aging/elderly screening reliability	Directly serves non-invasive/non-intrusive remote health monitoring + aging-in-place
Future NSTC potential	Strong basis for a follow-up proposal on "trustworthy AI for elderly cardiac screening," extensible to multimodal	Strong basis for a follow-up proposal on "personalized, calibration-light wearable vital-

Criteria	Student 1 — Project A	Student 2 — Project B
	(ECG+PPG) and to your own elderly cohort data later	sign monitoring for aging-in-place," extensible to real wearable pilot deployment

Ranking and Student Assignment

Both projects are ranked closely (1 and 2 by a small margin), because they are intentionally matched in feasibility and novelty — the ranking below is about student fit, not project quality.

Project B (Cuffless BP) is *marginally* more demanding technically (larger dataset engineering, signal-processing fiducial-point detection, a genuine two-dataset domain-shift design) — assign this to **the student with stronger data engineering / signal processing instincts or prior exposure to time-series regression**, since the personalization/fine-tuning loop and cross-dataset harmonization require more independent engineering judgment.

Project A (ECG) has a gentler on-ramp (single dataset, official splits already provided, well-documented conformal-prediction libraries) but a slightly more conceptually novel angle (trustworthy-AI auditing is a less mechanical, more "how do I define and interpret this" task) — assign this to **the student who is a strong Python/PyTorch programmer but may benefit from a dataset/pipeline that's easier to stand up quickly**, freeing more of their time to think carefully about the fairness/explainability analysis itself, which is where the real contribution lives.

If you have a clearer read on which student is stronger at signal-processing engineering vs. statistical/ML reasoning, that should be the deciding factor over this general guidance.

Topics Considered and Rejected (with reasons)

To follow your instruction to reject — not just omit — attractive but weak options:

PPG-based atrial fibrillation detection (e.g., MIMIC PERform AF subset / Simband-based studies).

Rejected as a *primary* project: most fully public, AF-labeled PPG datasets are small (dozens of subjects) and several widely-cited AF-PPG datasets used in the literature (e.g., Simband, UMMC) are not freely and immediately downloadable without direct author contact/DUA — this violates the "download quickly" requirement. Kept only as a **backup topic** below, using the PhysioNet-hosted MIMIC-III AF annotation subset, since that one is more directly accessible.

Sleep apnea / sleep staging from PPG/ECG using the Sleep Heart Health Study (SHHS) or MESA. Rejected as a *primary* project: these datasets, though excellent and richly labeled, are distributed through the **National Sleep Research Resource (NSRR)** and require a **data-use agreement and account approval**, which conflicts with your "should preferably not require a data-use agreement" and "start downloading quickly" constraints. Kept as a **backup**, using the fully open, no-registration PhysioNet **Apnea-ECG** database instead (smaller, 70 records, but zero access friction).

Wearable-based fall detection / elderly activity recognition (e.g., SisFall, MobiAct, UP-Fall). Rejected as a *primary* project: these are excellent, open, no-DUA datasets and genuinely well aligned with your gerontechnology interest, but they are **predominantly accelerometer/IMU-based**, not physiological-signal-

based (ECG/PPG/respiration/SpO₂), so they sit further from your stated primary research areas than Projects A and B. Kept as a **strong backup** given how directly it maps to "aging-in-place."

General "self-supervised pretraining foundation model for PPG" project. Rejected as *too ambitious* for a single TEEP student in 6–12 months: state-of-the-art physiological foundation models require large-scale multi-dataset pretraining and substantial compute, and evaluating "foundation model quality" rigorously is itself a research program, not a bounded student project. A **scoped-down version** (self-supervised pretraining on PPG-DaLiA + WESAD, fine-tuned for HR estimation) is offered as a backup, since prior work has already shown this specific scoped version is feasible and self-supervised pretraining measurably improves PPG-DaLiA heart-rate estimation.

Cuffless blood pressure estimation using the classic UCI "Cuff-Less Blood Pressure Estimation" dataset directly (i.e., without the leakage fix). Rejected as the primary approach specifically *because* of its subject-ID-free structure, which is exactly the flaw Project B is designed to correct — using it naively would reproduce a known, documented methodological error rather than contribute anything new.

Backup Topics (3–5, in case a primary dataset becomes unavailable)

PPG-based atrial fibrillation screening using the PhysioNet-hosted MIMIC-III AF annotation subset (ECG-based AF annotations linked to the MIMIC-III matched waveform subset) as a substitute classification task if PTB-XL access changes; smaller-scale but no-DUA and immediately downloadable.

Sleep apnea detection from single-lead ECG using the fully open PhysioNet **Apnea-ECG** database (70 records, per-minute apnea/non-apnea labels) — a safe fallback classification project requiring no data-use agreement.

Wearable-based elderly fall/activity recognition using **SisFall** or **MobiAct** (open, IMU-based, extremely fast to download and preprocess) — the most direct fallback for the "aging-in-place" priority if a physiological-signal project stalls on preprocessing difficulty.

Self-supervised representation learning for PPG heart-rate estimation, pretraining on WESAD + PPG-DaLiA and fine-tuning for HR estimation — a scoped-down version of the "foundation model" idea, directly buildable from datasets already verified as accessible in this report (both used already for Project B), so it is a low-friction pivot for either student if either primary project's main dataset becomes unavailable.

Multimodal stress/affect classification from ECG+EDA+respiration using **WESAD** (open, no DUA, 15 subjects, chest+wrist multimodal physiological signals with labeled stress/amusement/neutral states) — not directly cardiac-diagnostic, but solidly within "multimodal physiological signal analysis," with clean, immediately usable labels if a more clinical dataset proves harder to access than expected.

Note on file format: This report is provided as Markdown. If you'd like it reformatted as a Word document (e.g., for sharing with students or department administration), or split into two separate one-page project briefs for direct student handout, let me know and I can generate those as well.